



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP3744
ENTERPRISE SYSTEMS DESIGN & MODELLING
SECTION 2
GROUP 4

WRITING REPORT:

TITLE: ROBOTIC PROCESS AUTOMATION (RPA) IN ENTERPRISE SYSTEMS

LECTURER:

DR. NOORMINSHAH BINTI A. IAHAD

PREPARED BY:

EVELYN GOH YUAN QI (A23CS0222)
JOANNE CHING YIN XUAN (A23CS0227)

EXECUTIVE SUMMARY

This report discusses how Robotic Process Automation (RPA) enhances modern enterprise systems, such as Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), and Supply Chain Management (SCM). The objective is to analyze the technology's ability to improve efficiency, accuracy, and compliance in organizations that still rely heavily on manual and repetitive processes. The results of literature research and industry cases show that RPA can automate rule-based workflows, reduce human errors, reduce operating costs, and enable employees to focus on valuable work. Despite the significant advantages of RPA, its successful implementation still faces some challenges, such as poor governance structure, improper process design, and insufficient staff training. Case studies from companies such as Invesco, Siemens and OP Financial Group show that when RPA is effectively integrated, enterprises have achieved remarkable results in productivity and cost savings. The report concluded that organizations should adopt a phased and structured implementation method, taking into account human supervision while promoting technological innovation. In addition, it is recommended to gradually move towards Intelligent Automation (IA) to achieve more intelligent and data-driven enterprise operations by combining RPA with technologies such as Machine Learning (ML), Artificial Intelligence (AI), and Optical Character Recognition (OCR).

1.0 INTRODUCTION

In today's fast-changing digital world, most organizations are using enterprise systems such as Customer Relationship Management (CRM), Supply Chain Management (SCM), and Enterprise Resource Planning (ERP) as a way of streamlining their operations, decision making, and productivity. Nevertheless, these systems are associated with repetitive and rule-based work of employees, which decrease productivity and elevate the risk of human mistakes. Robotic Process Automation (RPA) can help solve these problems by using software robots known as bots that take up systemic actions like data entry and reporting. Powered by the same interfaces as the existing solutions without changes to the core architecture, RPA leads to improved speed, accuracy, and compliance, while allowing employees to perform more value-added operations (Durão & dos Reis, 2024). In addition, RPA is also a transition toward intelligent automation by introducing Artificial Intelligence (AI), Optical Character Recognition (OCR), and Machine Learning (ML) into the process execution in a smarter manner (Chennupati, 2025).

The purpose of this report is to present generalized information about Robotic Process Automation in Enterprise Systems, focusing on its technical functioning, strengths, drawbacks, issues, and organization trends. Besides, this report will showcase some real-world examples and give recommendations when there is a plan to use them effectively in enterprise systems.

2.0 BACKGROUND / LITERATURE REVIEW

Robotic Process Automation (RPA) has been attracting attention due to the fact that it allows organizations to automate simple tasks, allowing them to maintain their systems with minimal changes. Initial studies demonstrate that RPA is a digital person that automates human-like

tasks such as report generation, data migration, and invoice processing. As a result, personnel would be able to focus on more pertinent and tactical issues instead (Willcocks et al., 2015).

Flora et al. (2022) concluded that RPA usage has an important influence on the industries that use it. It leads to swift processing, higher precision, and better cost-effectiveness. Unlike traditional automation systems, RPA can work as a glue with the enterprise systems such as ERP, CRM, and SCM by emulating the way that users type, click, and access the items on their screens, and navigate. This makes it easy for enterprises still using slow and inefficient traditional systems (Fernandez & Aman, 2021). As such, bots are enabled to do simple decision-making, learning new patterns, and processing unstructured data (Siderska et al., 2023). In this way, organizations can automate entire workflows rather than individual tasks. To enable this, a second emerging trend is hyperautomation, which uses several automation tools, including cloud-based RPA platforms, to strategically drive full digital transformation across the enterprise systems (Durão & dos Reis, 2024).

These improvements enable automation to become more powerful, flexible, and intelligent. However, some fields still remain unexplored. Several articles focus on the efficiency of and organizational benefits of RPA, but they do not extensively cover ethical questions, the governance of data, and the effect on employees in the long term. Overall, the reviewed papers show that RPA is not only about increased enterprise efficiency but a crucial foundation step in automation transformation.

3.0 ANALYSIS AND DISCUSSION

3.1 HOW RPA WORKS IN ENTERPRISE SYSTEMS

RPA employs a digital automation technique where programs imitate human functionality for interacting with digital tools. It includes the following three main layers:

1. Bot Layer - Software-based robots that accomplish monotonous, rule-dependent jobs, such as inputting data or obtaining reports. The bot could either be attended, which involves human involvement in starting the automation process, or unattended, which is fully automated.
2. Control Room / Orchestrator - Offers a centralized platform that controls the bots with the assistance of scheduling and monitoring over various systems.
3. User Interface Layer - The way in which the bot connects with enterprise applications (ERP and CRM systems), undertaking screen-based tasks.

There are two types of RPA that are mainly used for integrating with enterprise systems. First, UI-based automation occurs, where robots complete jobs by recognizing things in the interface and duplicating keyboard and mouse activities. The second way is API-based automation, which is a way of interaction between enterprise applications like SAP or Oracle and bots without the help of visual UI. There are advanced models that integrate RPA with Artificial Intelligence and Optical Character Recognition (OCR), which involve analyzing unstructured data, while we have Machine Learning (ML) that need to facilitate decision-making (Chennupati, 2025). Since RPA manages existing systems the way they go without changing the inner code, it is a relatively painless and cheap method of process modification. It is particularly effective as it encompasses various legacy systems, which may be difficult to alter.

3.2 CASE EXAMPLES OR APPLICATIONS

Organization	Industry/ Domain	Enterprise Systems Involved	RPA platform	Key Outcomes
Invesco	Financial & Asset Management	Integrated financial management systems including compliance, accounting, and reporting modules	Blue Prism	Automated 165+ processes; annual savings of US \$2.1 million; reduced processing time; improved data accuracy and reporting efficiency (Blue Prism, n.d.)
OP Financial Group	Banking & Financial Services	Core banking infrastructure, customer relationship management, and back-office processing systems	Blue Prism	Automated 300+ processes; €40 million cost savings; reduced development time from 21 weeks to 9 weeks; improved compliance and service quality (Blue Prism, n.d.)
Siemens Mobility	Manufacturing & Transportation	ERP (SAP) platform and global operations management systems for procurement and logistics	UiPath	Scaled automation from 5 to 700+ processes; minimized manual workloads; improved SAP data integration and global productivity (Olavsrud, 2024)

Table 1: Real-world Examples of RPA in Enterprise Systems

3.3 ADVANTAGES AND DISADVANTAGES OF RPA IN ENTERPRISE SYSTEMS

Advantages	Disadvantages
Improved Efficiency and Productivity - RPA automates repetitive and rule-based processes, operating 24/7 to minimize processing time and increase throughput (Sobczak, 2022).	Implementation Complexity - Frequent updates to enterprise systems require continuous bot maintenance and configuration which increase operational workload (Altarazi et.al, 2024)
Enhanced Data Accuracy and Compliance - Bots execute predefined workflows consistently, reducing human error and improving data integrity (Durão & Reis, 2024).	Security and Governance Risks — Enterprise systems might be vulnerable if identity or role protection mechanisms, audit processes, or governance structures are deficient (Kedziora & Penttinen, 2021).
Non-Invasive Integration - RPA operates at the user interface level, allowing automation without altering core enterprise system code (Madakam et al., 2019).	Data & Integration Risks — Bot depend on clean, structured data and unchanging system interfaces; legacy system alterations or data quality problems can result in errors, workflow interruption, or maintenance demands (Dogan et al., 2024).

Table 2: Advantages and Disadvantages of RPA in Enterprise Systems

3.4 CHALLENGES AND FUTURE TRENDS

While presenting many benefits, its victory is also dependent on the processes that are effectively planned, monitored, and maintained by talented employees who use every smarter technologies. Robotization is widely adopted by numerous enterprises, yet implementing it in organized systems is complicated. Enterprises remain burdened with issues such as unstructured process documentation, weak integration of RPA and ERP modules, and bots that do not work on a consistent basis (Flora et al.,2022; Kraus et al.,2024). As well, employee resistance, insufficient training, and unreasonable return-on-investment expectations are usually organizational obstacles that prevent success (Kraus et al.,2024). At the same time, data privacy and safety issues also persist as automation tools access sensitive business data (Fernandez & Aman,2021).

In the future, AI-powered Intelligent Automation (IA) is gaining more ground by merging Artificial Intelligence and Machine Learning to allow self-operating, adaptive bots and to make a few simple decisions without human intervention (Siderska et al.,2023). In particular, hyperautomation, a future trend, joints together cognitive decision-making, analytics-driven workflows, and deep involvement in enterprise digital ecosystems (Palaniappan, 2024).

4.0 RECOMMENDATIONS

Employing a phased approach and combining organizational and technological components are critical to achieving Robotic Process Automation (RPA) in enterprise systems.

4.1 PROCESS IDENTIFICATION AND ALIGNMENT

Organizations should figure out which processes to target for automation. To find out which tasks should be repetitive and based on rules for automation, a comprehensive technical assessment should be performed, as live activities that rely on human logic and adaptation to a situation must be isolated for observation and control (Faizan, 2024). The alignment of IT solutions with business units to develop governing ecosystems to limit integration issues and improve performance (Kraus et al.,2024)

4.2 MANAGEMENT AND WORKFORCE ENABLEMENT

Organizations must have executive management changes and maintain periodical training informally on practical matters in order to boost employee morale and increase adaptability. Team members frequently face resistance and acceptance to automation when they do not seem to understand its goals and usefulness (Fernandez & Aman, 2021). Establishing a Centre of Excellence (CoE), should be part of structures in the organization to operate RPA practices and develop common processes of governance and follow data protection policies strictly (Kraus et al.,2024).

4.3 PERFORMANCE MEASUREMENT AND CONTINUOUS IMPROVEMENT

To assess the success of RPA in increasing performance and providing suggestions for further advances, enterprises should put together all the essential performance metrics to improve accuracy rates, cycle time, cost savings (Faizan, 2024) ,and feed the information to automation

experts. The most effective approach would be to integrate automation programs with a more comprehensive effort to speed up operational processes (Flora et al., 2022).

4.4 ADVANCEMENT TOWARD INTELLIGENT AUTOMATION

Lastly, enterprises should evolve their RPA systems and tools toward Intelligent Automation (IA) by integrating Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) to predict several outcomes and take more sensible decisions, thus allowing them to stay competitive in ever-changing enterprise surroundings (Siderska et al., 2023; Palaniappan, 2024).

5.0 CONCLUSION

RPA has become the main efficiency and digital transformation engine of the enterprise. RPA seeks to improve productivity, data quality, and focus employees on strategic tasks by automating repetitive, rule-based processes. RPA, when integrated with enterprise platforms like ERP, CRM, and SCM, leads to reduced operational expenses, increased scalability, and increased compliance. The organizations should develop management discipline structures in order to deal with the employee resistance, data security and absence of technical expertise.

Overall, RPA implementation must also be viewed as a comprehensive strategy of transformation, but not as a technical implementation solution. The organizations must balance the competitive spheres of implementation like technology, operations, workforce engagement, and corporate culture. In the future, RPA implementation with Artificial Intelligence (AI), Machine Learning (ML), and other cognitive technologies will result in Intelligent Automation (IA), that will enhance autonomy and flexibility of systems. Social implication of ethics, sustainable governance, hardships, and prospects of long-term working population are yet to be studied so that automation is evolved as to enhance the performance of the enterprise and human collaboration.

REFERENCES

- Altarazi, F., Santos, D., & Wong, E. (2024). Robotic Process Automation (RPA) Implementation Challenges: A Literature Review. In the 9th North American Conference on Industrial Engineering and Operations Management, <https://doi.org/10.46254/NA09.20240054>.
- Blue Prism. (n.d.). *Invesco doubles return on investment and saves \$2.1 million each year with Blue Prism*. Retrieved from <https://www.blueprism.com/resources/case-studies/invesco-doubles-return-on-investment-and-saves-2-1-million-each-year-with-blue-prism/>
- Blue Prism. (n.d.). *OP Financial Group – Longstanding innovation in enterprise automation*. Retrieved from <https://www.blueprism.com/resources/case-studies/op-fmea-longstanding-innovation/>
- Chennupati, N. (2025). Cognitive RPA: A framework for hybridizing artificial intelligence

- with robotic process automation in enterprise systems. *European Journal of Computer Science and Information Technology*, 13(19), 12–24 <https://ejournals.org/ejcsit/vol13-issue19-2025/cognitive-rpa-a-framework-for-hybridizing-artificial-intelligence-with-robotic-process-automation-in-enterprise-systems/>
- Dogan, O., Arslan, O., Cengiz Tirpan, E., & Cebi, S. (2024). Risk Assessment of Employing Digital Robots in Process Automation. *Systems*, 12(10), 428. <https://doi.org/10.3390/systems12100428>
- Durão, D., & dos Reis, A. P. (2024). How does robotic process automation create value for firms? *Information Systems and e-Business Management*, 22(4), 721–740. <https://link.springer.com/article/10.1007/s10257-024-00685-z>
- Durão, M., & dos Reis, J. (2024). *Enterprise automation and digital process integration using RPA tools*. *Journal of Digital Business Innovation*, 18(1), 23–34.
- Faizan, T. (2024). *Recommendations for using Robotic Process Automation (RPA) in a project reporting system* [Master's thesis, Metropolia University of Applied Sciences]
- Fernandez, D., & Aman, A. (2021). *The challenges of implementing robotic process automation in global business services*. *International Journal of Business and Society*, 22(3), 1269–1282. <https://doi.org/10.33736/ijbs.4301.2021>
- Flora, B., et al. (2022). *The impact of robotic process automation on operational efficiency in enterprises*. *Journal of Business and Management Research*, 22(3), 45–58.
- Kedziora, D., & Penttinen, E. (2021). Governance models for robotic process automation: The case of Nordea Bank. *Journal of Information Technology Teaching Cases*, 11(1), 20–29. https://www.researchgate.net/publication/343237905_Governance_models_for_robotic_process_automation_The_case_of_Nordea_Bank
- Kraus, P., Fißler, E., & Schlegel, D. (2024). A typology of challenges in the context of robotic process automation implementation projects. *International Journal of Organizational Analysis*, 32(11), 60–73. <https://doi.org/10.1108/IJOA-11-2023-4100>
- Madakam, S., Holmukhe, R. M., & Jaiswal, D. K. (2019). The future digital work force: Robotic Process Automation (RPA). *Journal of Information Systems and Technology Management*, 16(1), 1–17. <https://doi.org/10.4301/S1807-1775201916001>
- Olavsrud, T. (2024). Siemens Mobility scales RPA by empowering employees. CIO. Retrieved from <https://www.cio.com/article/2097016/siemens-mobility-scales-rpa-by-empowering-employees.html>
- Palaniappan, S. (2024). *An overview on robot process automation: Advancements, design*

standards, its application, and limitations. Journal of Emerging Technologies and Innovative Research, 11(5), 45–58.

Siderska, J., Aunimo, L., Süße, T., von Stamm, J., Kedziora, D., & Aini, S. N. B. M. (2023).

Towards intelligent automation (IA): Literature review on the evolution of robotic process automation (RPA), its challenges, and future trends. Engineering Management in Production and Services, 15(4), 90–103. <https://doi.org/10.2478/emj-2023-0030>

Sobczak, A. (2022). Robotic Process Automation as a Digital Transformation Tool:

classification of approaches and their consequences. *Sustainability, 14(3), 1333.* <https://doi.org/10.3390/su14031333>

Willcocks, L., Lacity, M., & Craig, A. (2015). *Robotic process automation: Strategic transformation lever for global business services.* The Outsourcing Unit Working Paper Series, 15/02.

Appendix / Logbook

Course: SECP3744 – Enterprise Systems Design & Modelling

Semester: 2025/2026 Semester I

Topic: Robotic Process Automation (RPA) in Enterprise Systems

Group: 4

Members: Evelyn Goh Yuan Qi (A23CS0222)

Joanne Ching Yin Xuan (A23CS0227)

1. Research and Writing Timeline

Date	Activity	Tools	Keyword	Details / Outcome
8 Oct 2025	Assignment briefing and topic selection	e-Learning, Google Sheets	“Enterprise Systems”, “RPA”, “digital transformation”	After the lecturer’s briefing, our group decided to choose “Robotic Process Automation (RPA) in Enterprise Systems”. Google Sheets used to record members and topics
10-11 Oct 2025	Task division & Report planning	Spin the Wheel (online), e-Learning	“RPA report structure,” “task division,” “team collaboration”	Task distribution was completed on 10 October 2025, and the group agreed on deadlines for drafting, review, and final submission. E-Learning was used to check the rubric, and Spin the Wheel was used to assign tasks. Final task list: • Executive Summary – Joanne • Introduction – Evelyn • Literature Review – Joanne • Analysis and Discussion – Evelyn • Challenges/Future Trends – Joanne • Recommendations – Joanne • Conclusion – Evelyn.

12–15 Oct 2025	Research on RPA basics	Google Scholar, IEEE Xplore, ChatGPT, e-Learning	“Robotic Process Automation (RPA)”, “Intelligent Automation (IA)”, “ERP systems”, “automation challenges”, “RPA benefits”, “case studies”, “recommendations”	Collected resources related to the topic. Used ChatGPT with the prompt: <i>“Summarize RPA research findings and organize them by benefits, challenges, and future trends in point form.”</i> Reviewed the Technical Report template and rubric via e-Learning. Planned the report structure, identified key focus areas, and confirmed the inclusion of visuals and APA 7th citation style.
16 Oct 2025	Introduction	Google Scholar, ChatGPT, Microsoft Word	“Enterprise systems”, “RPA importance”, “digital transformation”, “report objectives”	Developed the introduction to provide background on enterprise systems and explain the importance of adopting RPA for efficiency and digital transformation. Used ChatGPT to help structure the flow of ideas. Prompt used: <i>“Give a short structure for an introduction about RPA in enterprise systems.”</i> All AI-generated results were paraphrased and refined manually before inclusion in the Microsoft Word document.
17 Oct 2025	Background/ Literature review	Google Scholar, ChatGPT, Microsoft Word	“Robotic Process Automation (RPA)”, “automation benefits”, “automation challenges”, “enterprise systems”, “Intelligent Automation (IA)”, “AI and ML integration”, “hyperautomation”, “digital transformation”	Summarized the main findings on RPA benefits and challenges from selected journal articles. Used ChatGPT with the prompt: <i>“Summarize advantages and challenges of RPA from academic sources.”</i> The AI output was paraphrased and reviewed against sources to ensure factual accuracy and citation consistency before inclusion in the report.
18–19 Oct 2025	Analysis and Discussion	Google Scholar, Blue Prism official site, CIO, AI ChatGPT, Microsoft Word	“RPA architecture”, “ERP and CRM integration”, “case studies”, “advantages”, “disadvantages”, “challenges”, “future trends”, “Intelligent Automation (IA)”, “hyperautomation”	A. Technology works or functions: Prompt used: <i>“Explain how RPA works including bot components, architecture and ERP/CRM integration in enterprise systems.”</i> B. Case examples or applications: Prompt used: <i>“Provide real industry case examples of RPA usage with clear business benefits in enterprise environments.”</i>

				C. Advantages and disadvantages: Prompt used: “List advantages and disadvantages of RPA in enterprise systems with short explanations.” D. Challenges and future trends: Prompt: “<i>Summarize RPA challenges and future trends</i>”. All AI-generated summaries were cross-checked to ensure factual accuracy and consistency before inclusion in the final report.
20 Oct 2025	Recommendations	Microsoft Word, ChatGPT, Google Scholar	“RPA”, “Intelligent Automation (IA)”, “implementation”, “training”, “governance”, “AI and ML”, “digital transformation”	Used ChatGPT with the prompt: “ <i>Summarize key recommendations for implementing RPA in enterprise systems.</i> ” Verified the output with research sources. Finalized points on phased adoption, staff training, and transition to Intelligent Automation (IA).
21 Oct 2025	Conclusion	Microsoft Word, ChatGPT	“Report findings”, “RPA effectiveness”, “enterprise systems”, “future automation”	Summarized the key findings of the report, highlighting how RPA improves enterprise efficiency and accuracy. Added final points on governance, employee training and future transition toward Intelligent Automation. Prompt used: “ <i>Summarize the key findings of RPA in enterprise systems.</i> ”
22 Oct 2025	AI verification & Create a detailed appendix/logbook section, including AI prompts and verification steps.	Grammarly, Turnitin, Microsoft Word	“AI verification”, “grammar check”, “logbook preparation”, “academic writing”, “report clarity”	Used Grammarly to refine grammar and Turnitin to check originality and improve clarity. Used ChatGPT to structure and summarize the logbook format. All sections were manually reviewed before inclusion in the final report.
23–25 Oct 2025	Editing and final review	Microsoft Word, Grammarly, Turnitin,	“APA citation”, “report formatting”, “final editing”, “proofreading”, “submission”	Check all sections, formatted headings and tables, reviewed APA citations, and performed final grammar and originality checks. Exported and finalized the report in PDF format for submission.

2. Verification Steps

Each section of the report was thoroughly verified to ensure accuracy and originality. All AI-generated summaries, tables, or paraphrased content were checked against the original journal sources before inclusion in the final document. No AI-generated text was used without manual editing and review to maintain academic integrity. Every reference was cross-checked through DOI links and formatted according to the APA 7th edition guidelines. Finally, Grammarly was used for grammar and clarity improvement, while Turnitin was employed to verify originality and ensure that the final report met academic standards.

3. Reflection

This assignment enhanced our understanding of RPA as a digital transformation tool in enterprise systems. Using AI tools such as ChatGPT supported the report structure, paraphrasing, and clarity, but not content creation. All outputs were verified to maintain academic integrity. The experience also improved our skills in collaborative research, responsible AI use, and professional technical writing.